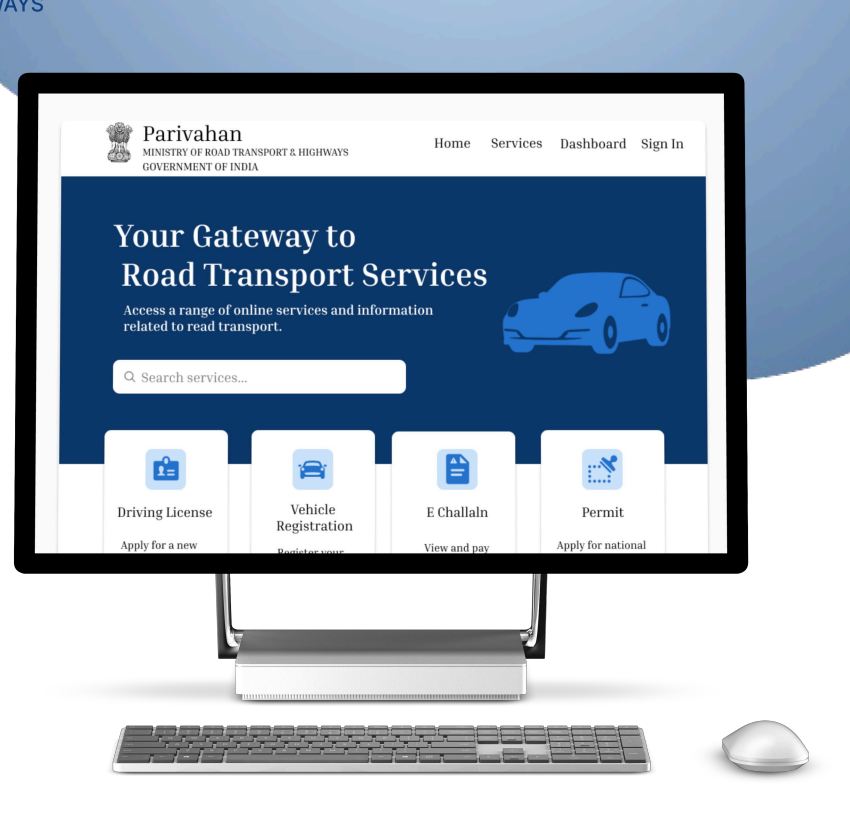
PARIVAHAN
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
GOVERNMENT OF INDIA

UI/UX
CASE STUDY

Parivahan is the national transport service portal. This case study focuses on redesigning its digital experience to make services like Driving License, Vehicle Registration, and e-Challan more intuitive, faster, and user-friendly for Indian citizens.


Parivahan
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
GOVERNMENT OF INDIA
Home
Services
Dashboard
Sign In
Your Gateway to Road Transport Services
Search services...
Driving License
Vehicle Registration
E Challan
Permit

Project Overview

The Parivahan portal, managed by the Ministry of Road Transport & Highways, offers essential services like driving license application, vehicle registration, eChallan payment, and permits.
However, its outdated interface, cluttered layout, and non-responsive design make navigation difficult for users.
This redesign project focuses on creating a modern, user-centric experience that is:

- Visually appealing
- Easy to navigate
- Mobile-responsive
- Inclusive of all core services with streamlined access

Duration:
- 6 weeks

Screens:
10+ screens

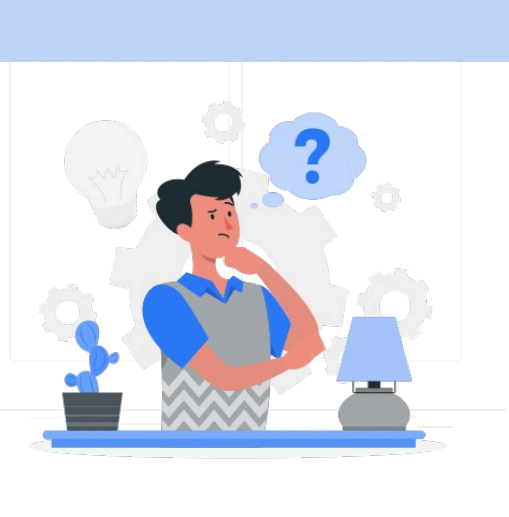
Tools:

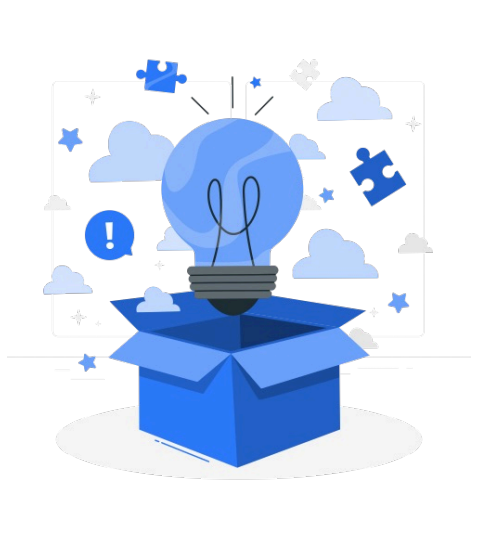
My Role

Design Strategy
Problem Solution
Information Architecture
Empathy Mapping
Usability Testing
User Flow
Prototyping
Wireframes
Competitive Analysis
Visual Design
User Research
User Persona

Problem Statement

The current Parivahan portal suffers from outdated design, poor mobile responsiveness, and complex navigation. Users struggle to locate services like license status or vehicle registration. The cluttered layout and lack of visual hierarchy lead to user frustration, incomplete tasks, and reduced trust in digital government services.





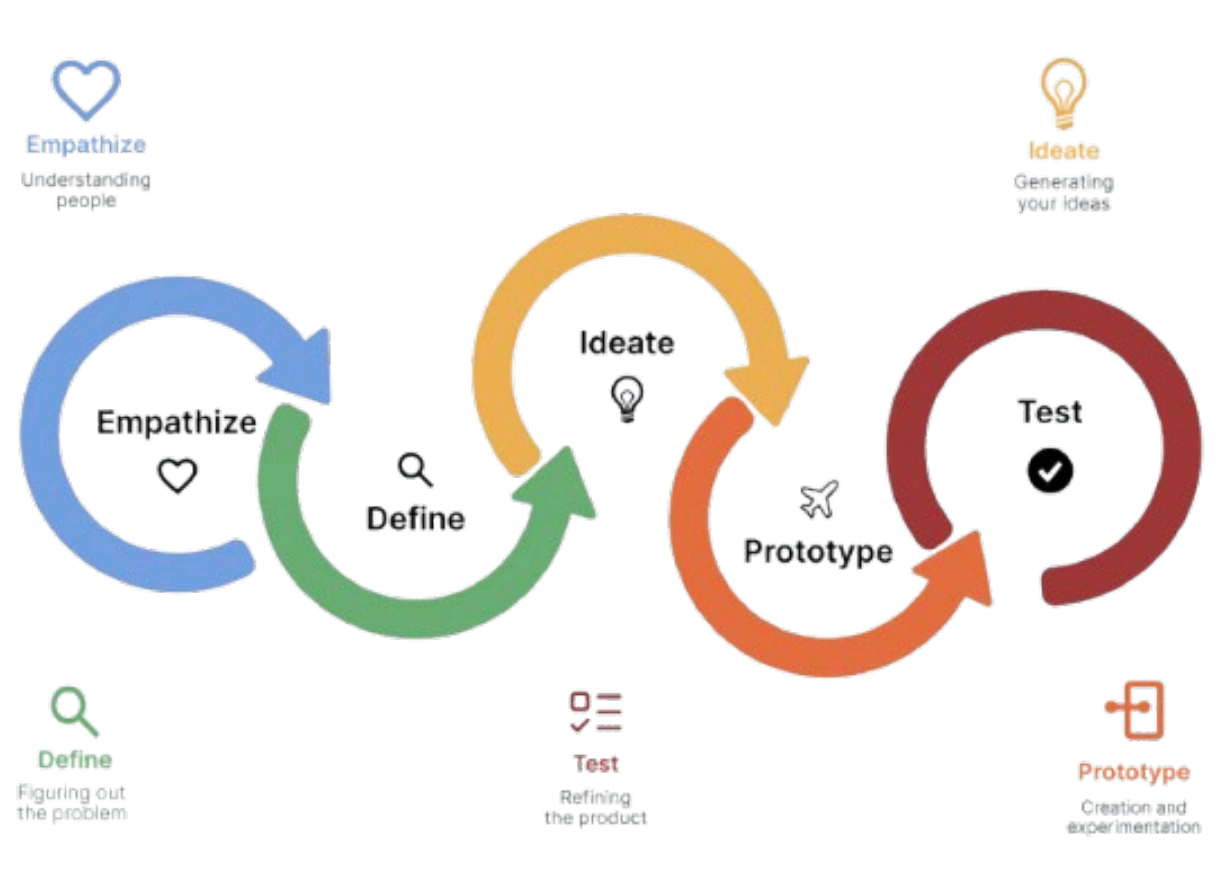
Problem Statement

The redesigned Parivahan portal provides a clean, intuitive, and mobile-responsive interface. It simplifies service access through structured menus, visual hierarchy, and clear CTAs. User-centered design enhances task completion, improves accessibility, and reduces dependency on offline agents, making transport services easier and more efficient for all.

Target Audience

- General public using transport services
- First-time driving license applicants
- Vehicle owners and dealers
- Government transport department officials

Design Thinking Process


Empathize
Understanding people
Ideate
Generating your ideas
Empathize
Define
Ideate
Prototype
Test
Define
Figuring out the problem
Test
Refining the product
Prototype
Creation and experimentation

01

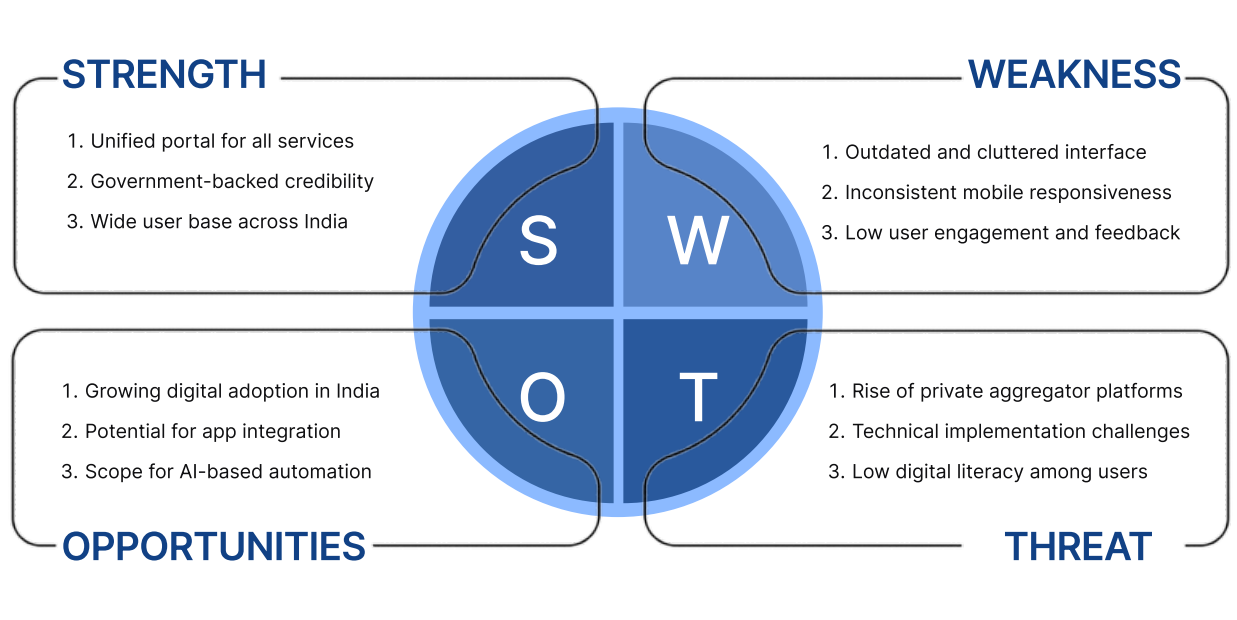
Empathize Phase

Market research
User research
Competitive analysis
SWOT analysis

Competitor Analysis

Feature/ App	mParivahan App	RTO Citizen App	Sarathi Portal
Key Features	Virtual DL/RC, eChallan status	DL application, vehicle info	DL services, appointment booking
Strengths	Easy access to documents	Localized features per state	Focused on licensing workflows
Weaknesses	Limited service coverage	Inconsistent UI, outdated design	Poor mobile responsiveness

SWOT analysis



STRENGTH
1. Unified portal for all services
2. Government-backed credibility
3. Wide user base across India

WEAKNESS
1. Outdated and cluttered interface
2. Inconsistent mobile responsiveness
3. Low user engagement and feedback

OPPORTUNITIES
1. Growing digital adoption in India
2. Potential for app integration
3. Scope for AI-based automation

THREAT
1. Rise of private aggregator platforms
2. Technical implementation challenges
3. Low digital literacy among users

Market Research

- **User Demand:** High need for digital transport services.
- **User Trends:** Preference for mobile-first, easy UI.
- **Competitors:** UMANG, mParivahan, DigiLocker simplify access.
- **Pain Points:** Confusing layout, slow mobile UX.
- **Opportunity:** Improve usability with modern design.

User Research

User research involved observing how different user groups—like students, vehicle owners, and license applicants—interacted with the existing Parivahan portal. Most users found it difficult to navigate, especially on mobile. The cluttered interface and deep menu structures caused confusion. These insights helped us identify the core needs of users and guided our redesign to focus on accessibility and simplicity.

Needs Identified:

- Quick access to core services
  Reduce time spent searching key features.
- Simplified and clear navigation
  Intuitive paths to complete common tasks.
- Responsive mobile-first interface
  Seamless use on various screen sizes.
- Minimal visual and cognitive load
  Cleaner design with reduced clutter and options.

02

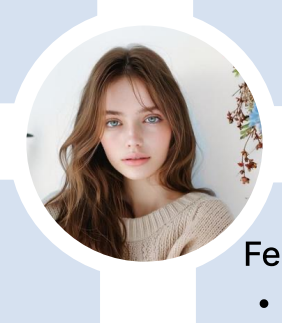
Define Phase

Empathy Map
Persona
Story Board
Project Goals

Empathy Map

Says
- "The website is too confusing to use."
- "I can't find the service I need quickly."

Thinks
- "This should be simpler like other modern apps."
- "I hope I don't make a mistake filling the form."



Does
- Keeps switching tabs or opens YouTube tutorials for help.
- Relies on cyber cafés or agents to complete tasks.

Feels
- Frustrated with the slow and outdated interface.
- Overwhelmed by the number of unclear service options.

Persona

Vanya Gupta

Age	24
Status	Single
Occupation	HR Executive
Location	Mumbai, India

Personality Traits

Tech-aware
Organised
Impatient

Background

Vanya Gupta is a 25-year-old HR executive working in a fast-paced corporate office in Mumbai. She juggles multiple responsibilities and prefers quick, digital solutions for errands. With limited time during workdays, she expects government services to be mobile-friendly, efficient, and transparent. She values user-centric design and often compares public platforms with private apps.

Goals
- To complete tasks like license renewal and vehicle registration independently.
- To find correct information quickly without external help.
- To access services from her mobile without visiting physical offices.

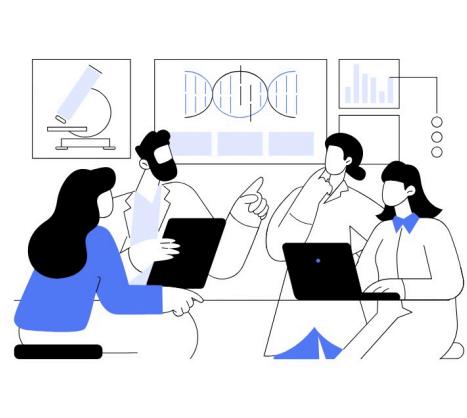
Frustrations
- Too many unclear links and outdated layouts on official websites.
- Long wait times and lack of confirmation or status tracking.

Needs
- Clear and intuitive navigation.
- Step-by-step guidance for every service.
- Mobile-responsive interface.
- Real-time updates on application status.

Motivations
- Wants to avoid taking leave or standing in queues.
- Prefers digital tools that save time and effort.
- Feels empowered when she can do things independently online.

Story Board

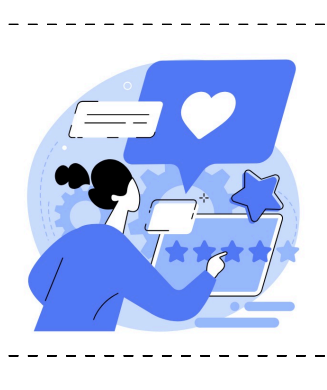
Vanya scrolls through messy government sites, feeling frustrated again.



She hears about the new Parivahan app from colleagues.

Vanya installs the app and sees a clean dashboard.

She selects "Renew License" and uploads documents within minutes.



App sends status updates; Riya relaxes with coffee at desk.

License approved! Riya smiles, recommends app to her work group.

Project Goals

- **Simplify Navigation**
  Make information and services easily accessible through an intuitive interface.
- **Enhance User Experience**
  Provide a clean, responsive, and visually organized platform for all users.
- **Streamline Core Services**
  Ensure quick access to high-demand features like license renewal and vehicle registration.
- **Improve Trust and Transparency**
  Offer real-time status tracking and clear instructions at every step.
- **Reduce User Frustration**
  Minimize waiting times, broken links, and confusing form submissions.
- **Mobile-First Accessibility**
  Ensure the platform is optimized for smartphones, on-the-go usage, and lower bandwidth.

03

Ideate Phase

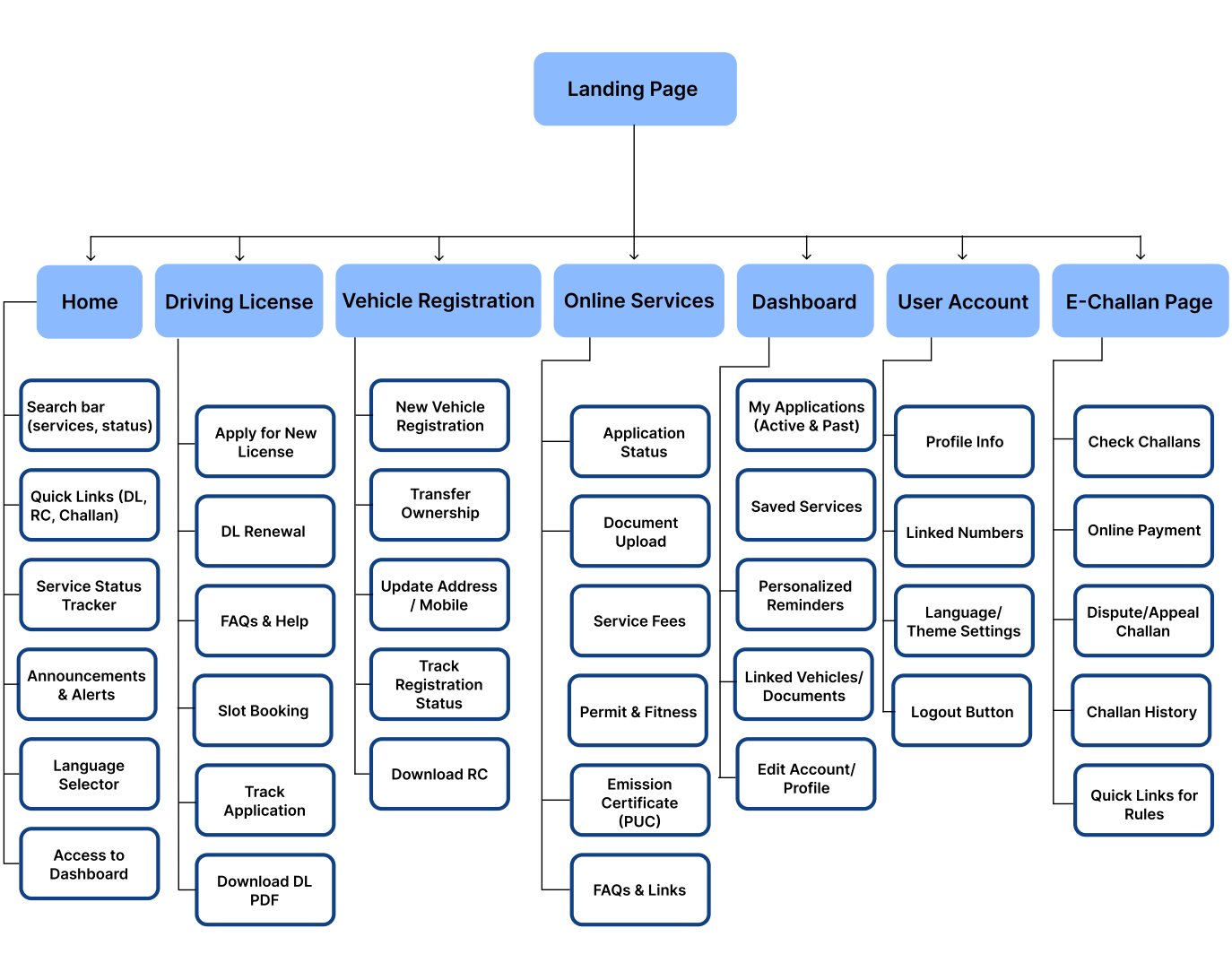
Why are we different>?
Information Architecture
User Flow

Why are we different?

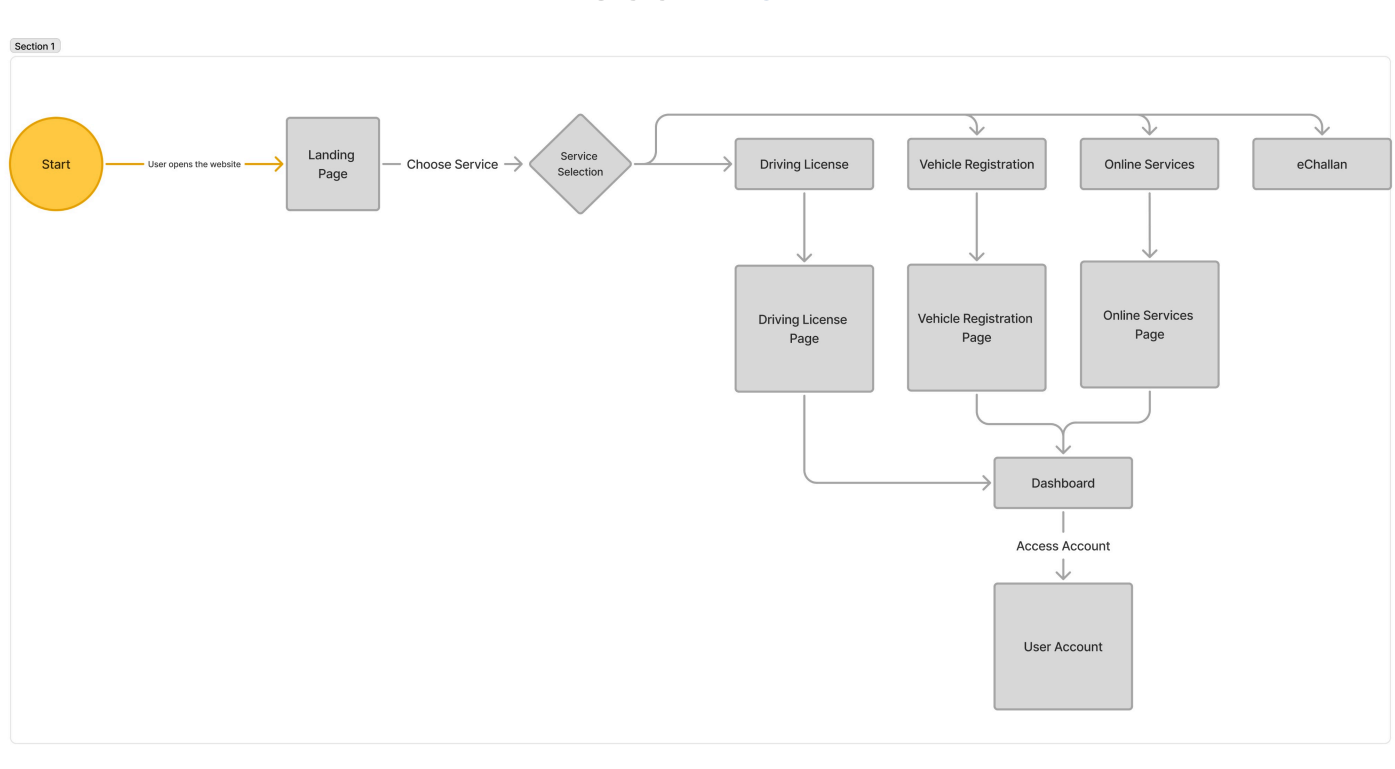
- User-Centered Design
  Our interface is designed based on real user pain points and feedback.
- Simplified Navigation
  Unlike the cluttered current site, we provide clear pathways to key services.

- Quick Access to Essentials
  Smart shortcuts for common actions like license renewal and challan payment.
- Mobile-First Experience
  Fully responsive design tailored for users on the go.
- Status Transparency
  Live updates and progress tracking for applications and verifications.
- Visual Clarity & Accessibility
  Minimalist layout, regional language support, and accessibility-friendly UI.

Information Architecture


Landing Page
Home
Search bar (services, status)
Quick Links (DL, RC, Challan)
Service Status Tracker
Announcements & Alerts
Language Selector
Access to Dashboard
Driving License
Apply for New License
DL Renewal
FAQs & Help
Slot Booking
Track Application
Download DL PDF
Vehicle Registration
New Vehicle Registration
Transfer Ownership
Update Address / Mobile
Track Registration Status
Download RC
Online Services
Application Status
Document Upload
Service Fees
Permit & Fitness
Emission Certificate (PUC)
FAQs & Links
Dashboard
My Applications (Active & Past)
Saved Services
Personalized Reminders
Linked Vehicles/ Documents
Edit Account/ Profile
User Account
Profile Info
Linked Numbers
Language/ Theme Settings
Logout Button
E-Challan Page
Check Challans
Online Payment
Dispute/Appeal Challan
Challan History
Quick Links for Rules

User Flow


Section 1
Start
User opens the website
Landing Page
Choose Service
Service Selection
Driving License
Vehicle Registration
Online Services
eChallan
Driving License Page
Vehicle Registration Page
Online Services Page
Dashboard
Access Account
User Account

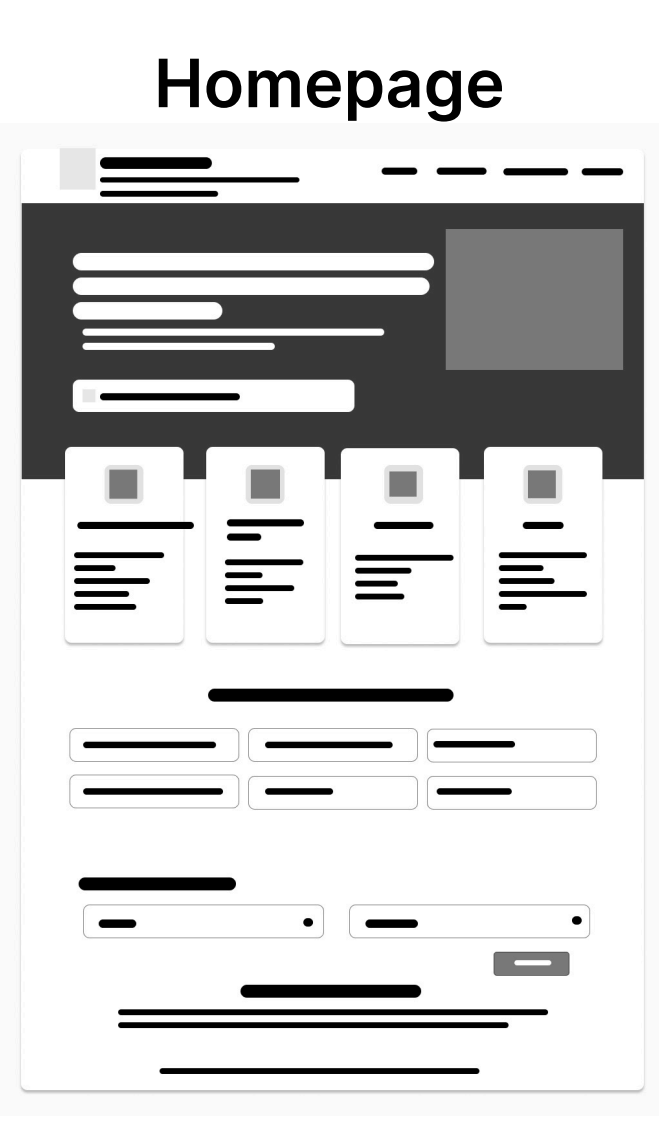
04

Design Phase

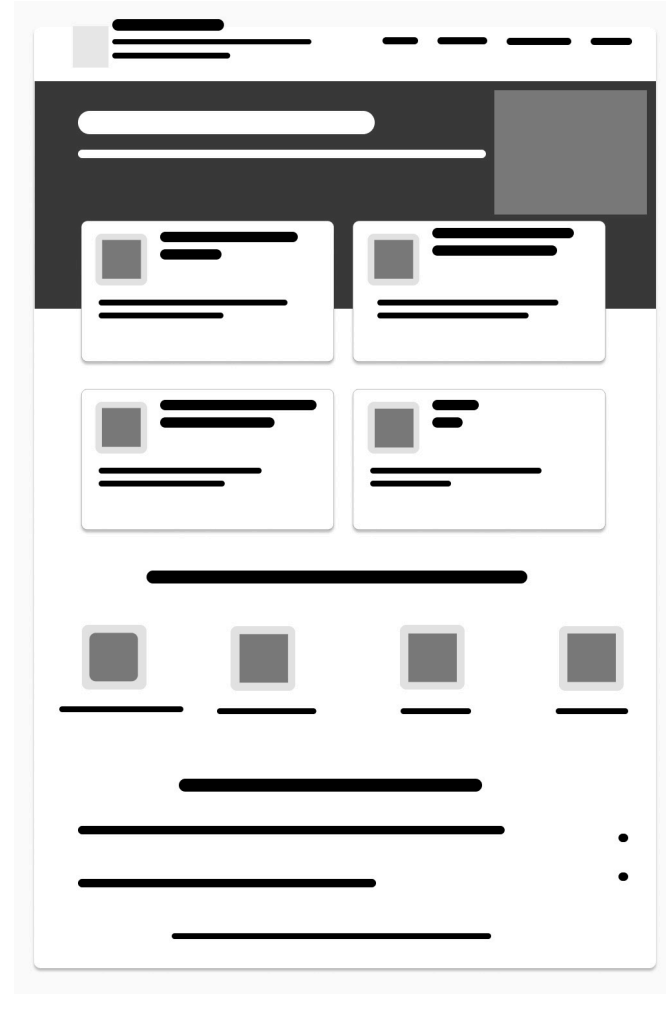
Wireframes
Typography
Ui Design

Mid-Fidelity Wireframe

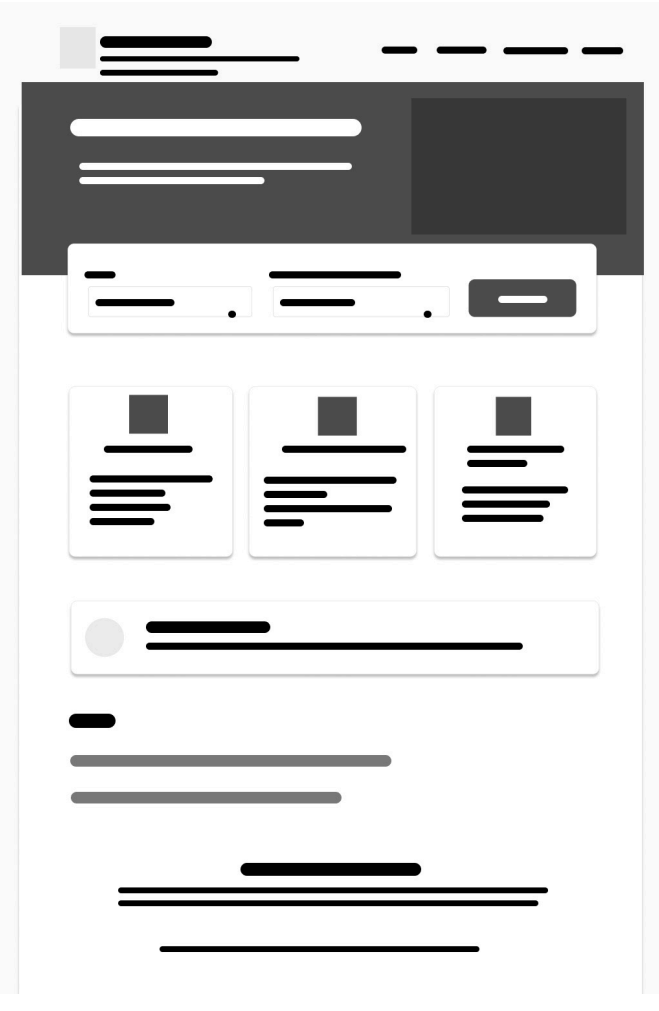
Homepage



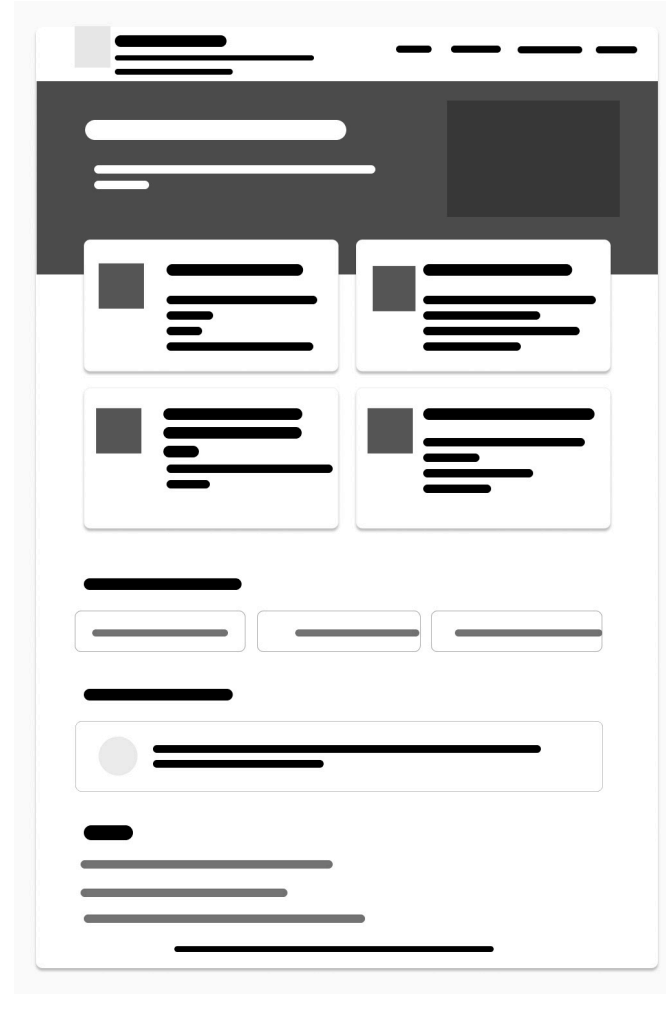
DL Page



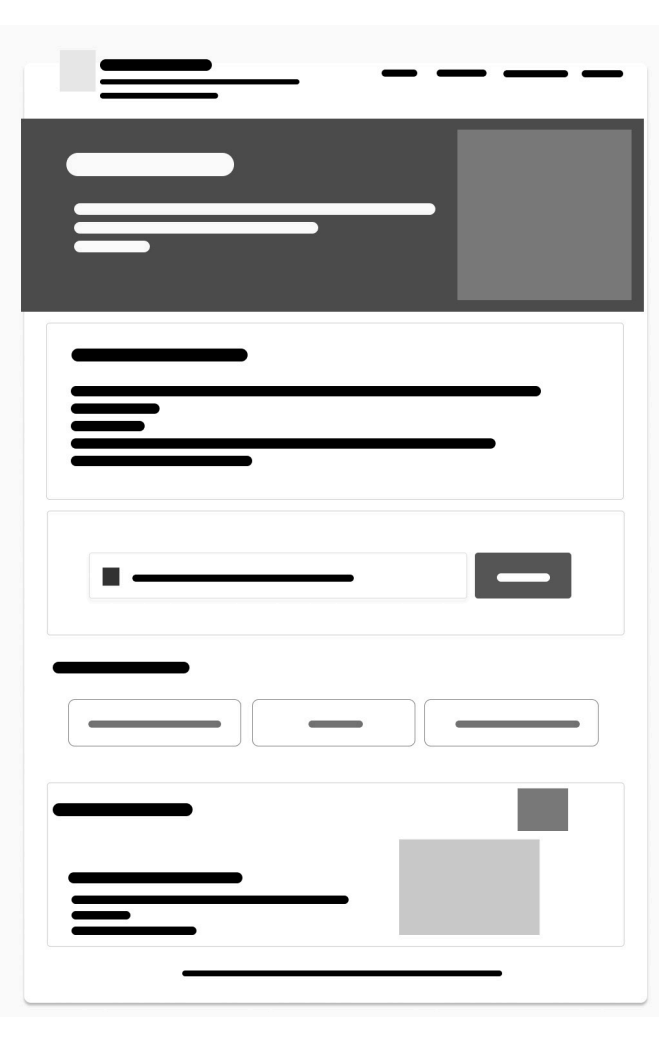
Vehicle Registration Page



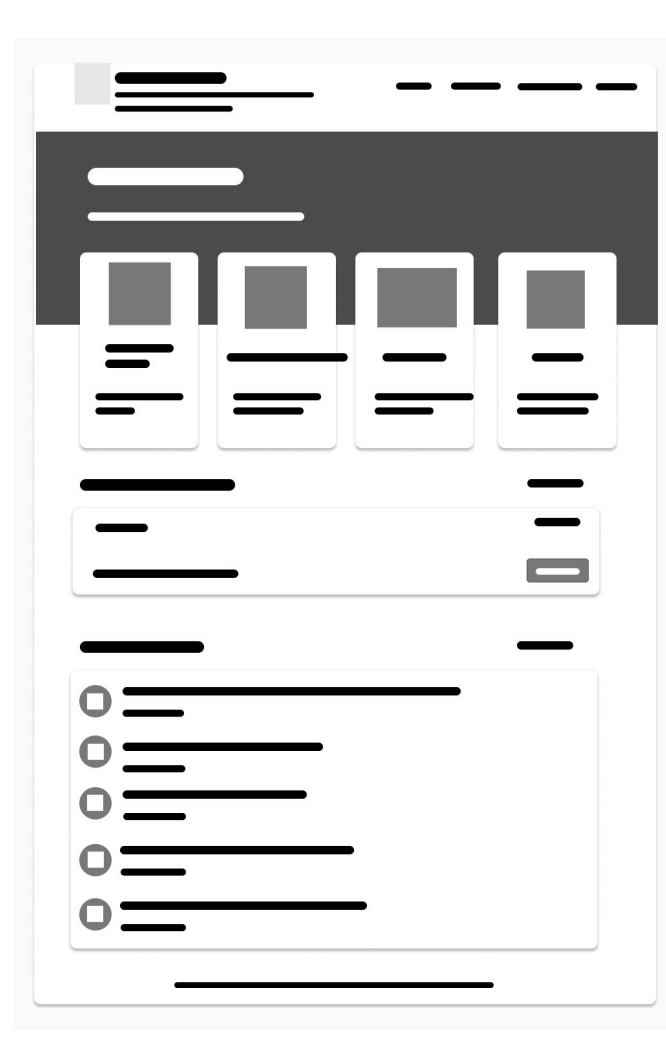
Online Services Page



eChallan Page



Dashboard Page



Typograpghy

Aa

Inria Serif

Regular Medium **Semibold** **Bold**

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

a b c d e f g h i j k l m n o p q r s t u v w x y z

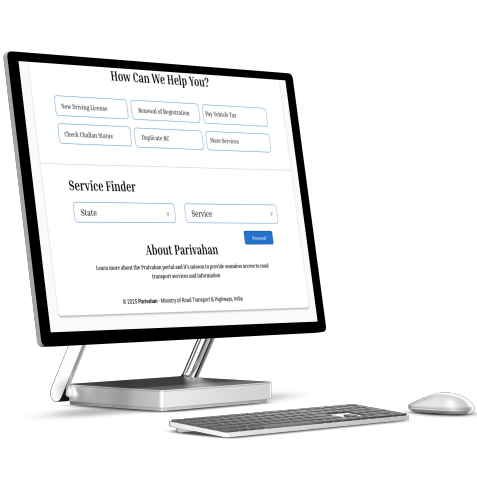
0 1 2 3 4 5 6 7 8 9

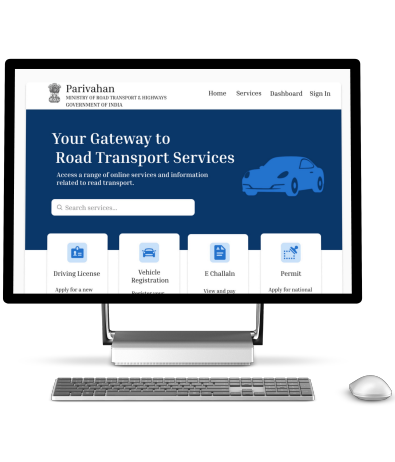
Color Palette

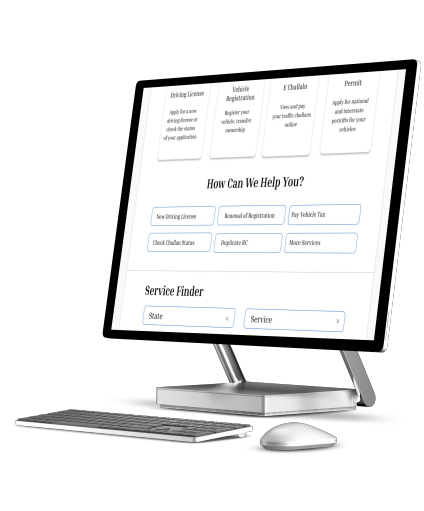
000000
000000
2473CF
CBE2FA
FFFFFF
0C3769
FBFAFB
E31514

UI Design

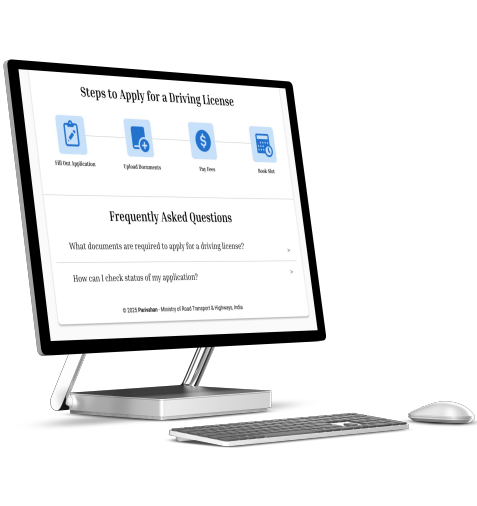
Homepage Screen

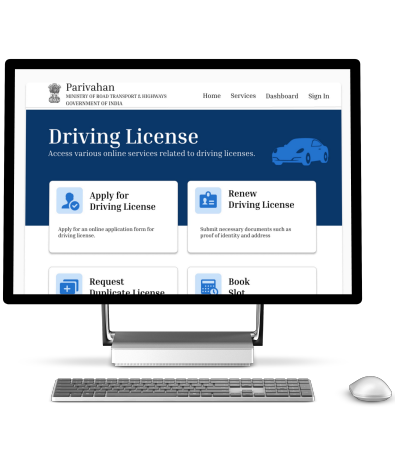


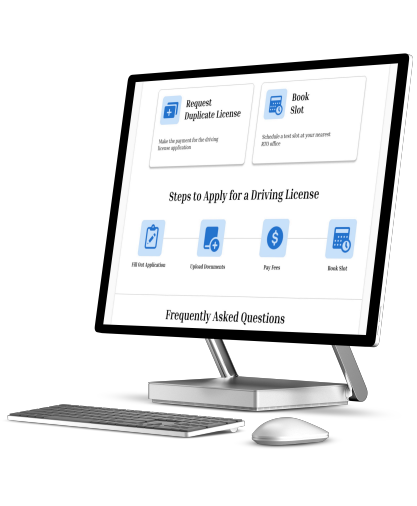




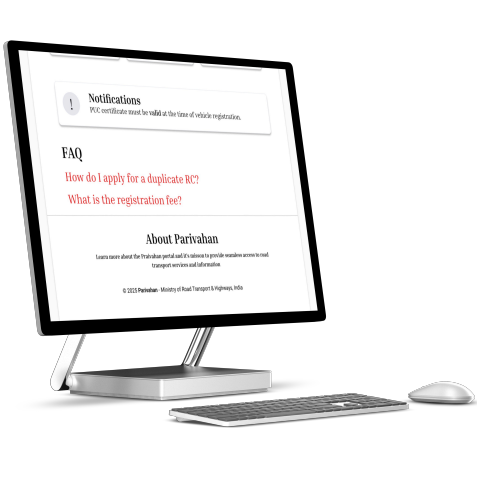
Driving License Screen

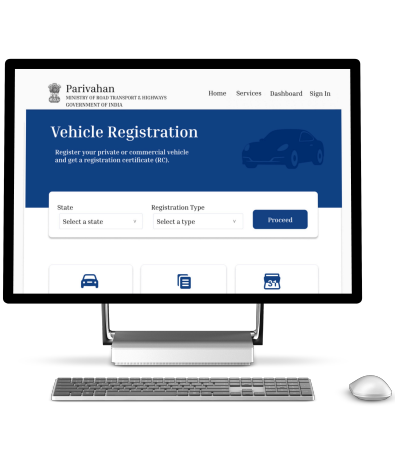


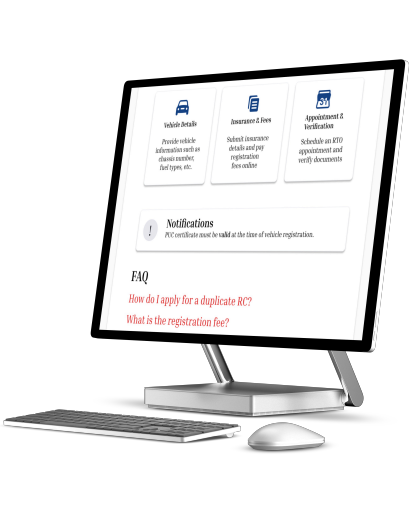




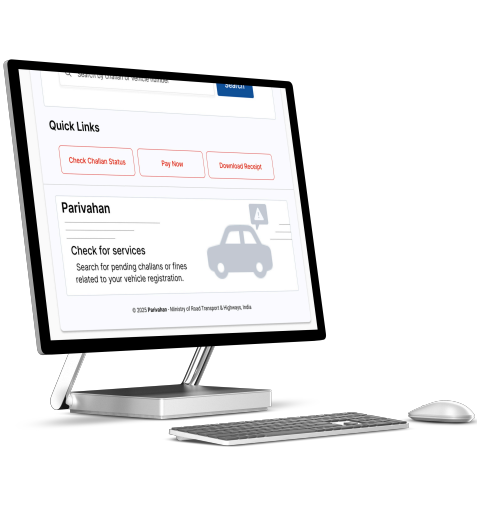
Vehicle Registration Screen

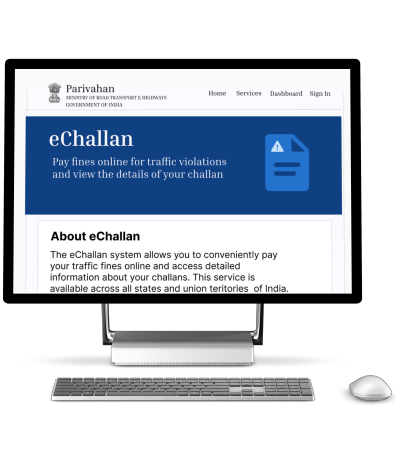


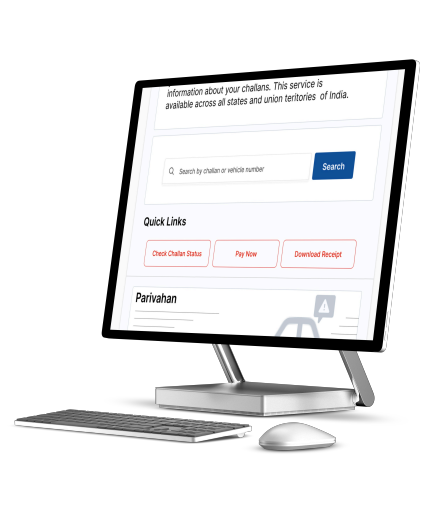




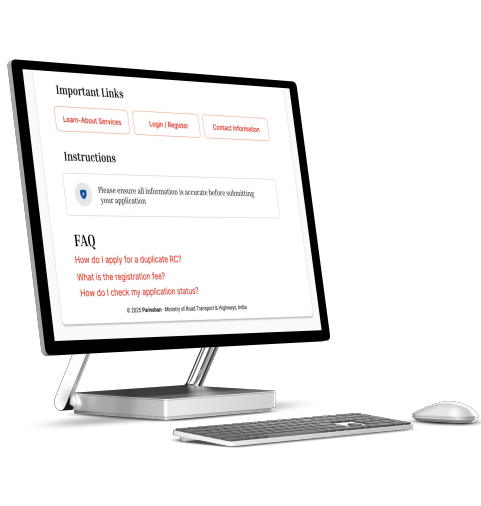
eChallan Screen

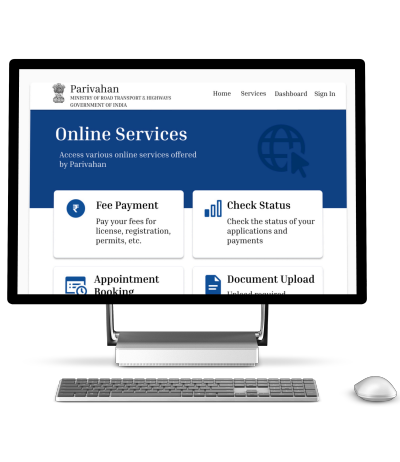


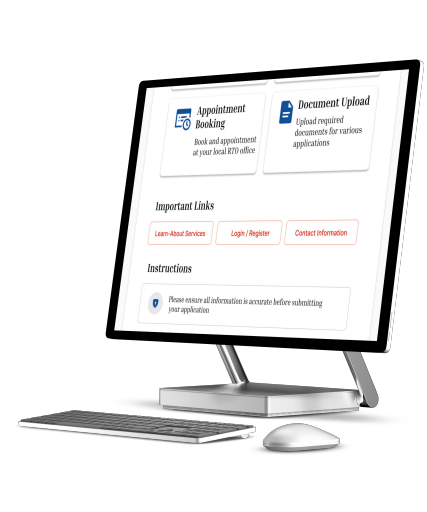




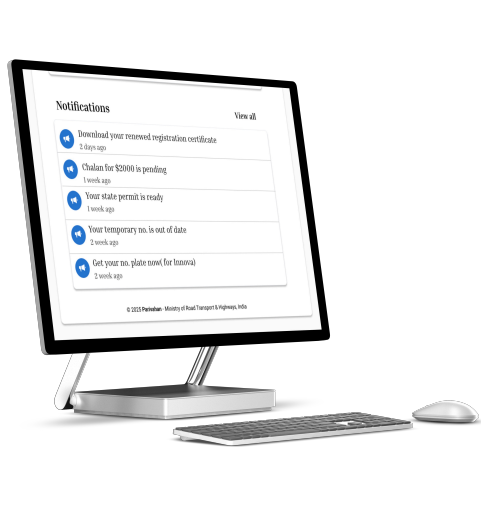
Online Services Screen

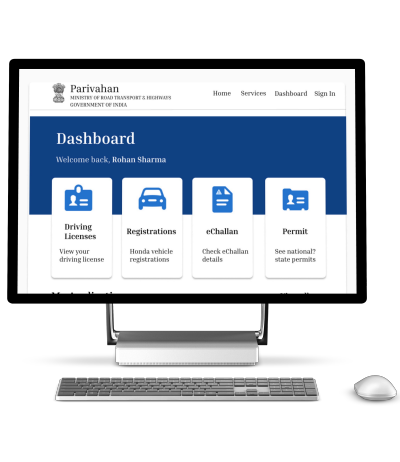


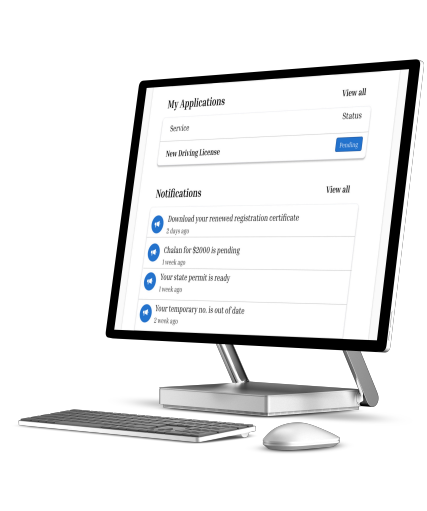




Dashboard Screen







Thanks for Watching

have you love it!

Do view the designs!
Also give your feedback!

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